

EXP NO 11

Implement an application that writes data to the SD card

AIM:

To develop an Android application using Android Studio that allows users to enter data and store it in the device's SD card (external storage).

ALGORITHM:

1. Input:

- ☐ The user enters text data in the application.
- ☐ The user clicks a button to save the data into the SD card.

2. Steps for Implementation in Android Studio:

1. Create a New Android Project:

- ☐ Open Android Studio
- ☐ Create a new project with an Empty Activity
- ☐ Choose Java or Kotlin as the programming language

2. Design the User Interface:

- ☐ Add the following components in the layout:
 - o EditText → to enter data
 - o Button → to save data
 - o TextView (optional) → to display status messages

3. Add Permissions in Manifest File:

- ☐ Declare permission to write to external storage
- ☐ For modern Android versions, also handle runtime permissions

4. Handle Runtime Permissions:

- ☐ Check if permission is granted
- ☐ If not, request permission from the user at runtime
- ☐ Proceed only after permission is granted

5. Check External Storage Availability:

- ☐ Ensure that:
 - o SD card is available
 - o SD card is writable

6. Create File in SD Card:

- ☐ Define a file name (e.g., mydata.txt)

- ☐ Select appropriate directory:

- o App-specific folder (recommended)

- o Or public directory

7. Write Data to SD Card:

- ☐ Read input from EditText

- ☐ Open file output stream

- ☐ Write the data into the file

- ☐ close the stream

8. Display Output to User:

- ☐ Show confirmation message:

- o “Data saved successfully”

- o Or error message if something fails

9. Handle Exceptions:

- ☐ Manage:

- o IOException

- o Permission denial

- o Storage unavailability

3. Testing the Application:

- ☐ Run the app on a real device/emulator

- ☐ Enter sample text

- ☐ Click Save button

- ☐ Verify file using file manager in external storage

4. Observations:

- ☐ Data is stored successfully in the SD card

- ☐ Runtime permission handling is essential

- ☐ External storage path must be handled carefully in newer Android versions

OUTPUT:

- ☐ User enters text → Clicks Save

- ☐ Message displayed:

Data saved successfully to SD card

- ☐ File mydata.txt created in external storage containing user input

CODE:

ACTIVITY_MAIN.XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

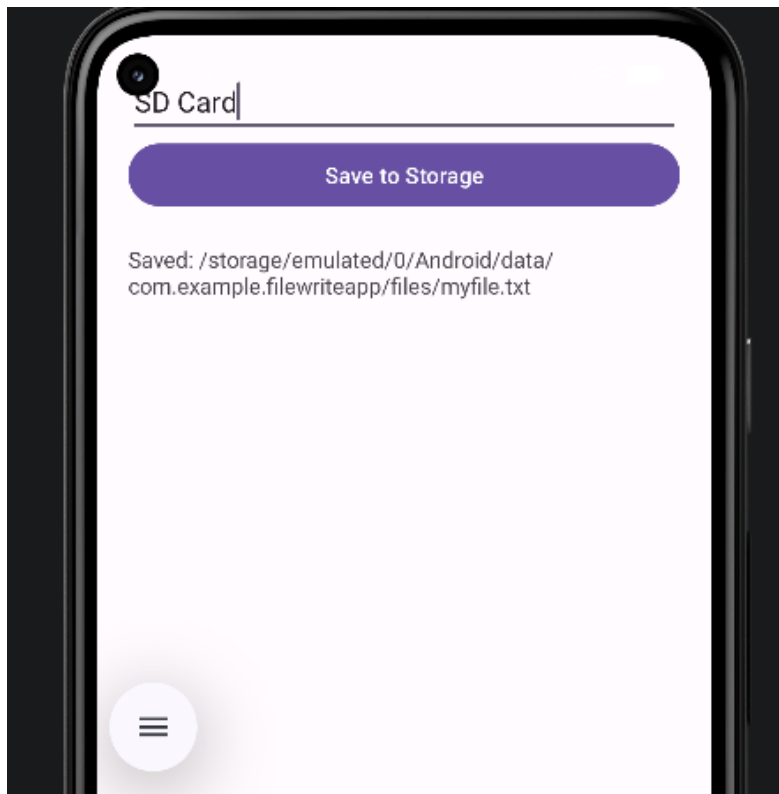
    <EditText
        android:id="@+id/inputText"
        android:hint="Enter text to save"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/saveBtn"
        android:text="Save to Storage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/status"
        android:text=""
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

OUTPUT:



RESULT:

An Android application is successfully developed using Android Studio to write data to the SD card with proper permission handling and storage management.